

FastAPI + MongoDB Lab

- Build a REST API using FastAPI, MongoDB, and Docker (Linux workflow)

Step 1: Create Project

```
# create project folder  
mkdir fastapi-mongo-lab
```

```
#move to the folder  
cd fastapi-mongo-lab
```

Step 2: Virtual Environment

#create env

```
python3 -m venv venv
```

#activate env

```
source venv/bin/activate
```

#upgrade pip

```
python -m pip install --upgrade pip
```

Step 3: Install Dependencies

```
#install dependencies
```

```
pip install fastapi uvicorn motor pydantic
```

```
#check installation
```

```
uvicorn --version
```

Step 4: Docker Compose File

#create a docker-compose.yml
nano docker-compose.yml

#copy paste the following text

- version: '3.8'
- services:
- mongo:
- image: mongo
- ports:
 - '27017:27017'

#press ctrl and letter o to save it and press enter
ctrl+ o

#press ctrl+x to close nano
Ctrl + x

Step 5: Run MongoDB

```
# run MongoDB
```

```
docker compose up -d
```

```
# make check
```

```
docker ps
```

Step 6: Create FastAPI code

- Define Student model using Pydantic

in a new cmd open new nano
nano models.py

Step 6: Create FastAPI code

#write the code

```
from pydantic import BaseModel
```

```
class Student(BaseModel):
```

```
    student_id: str
```

```
    name: str
```

```
    grade: int
```


Step 7: Create main.py

- Setup FastAPI app and MongoDB connection
- Define CRUD endpoints

Step 7: Create main.py

```
#open nano  
nano main.py
```

#write

- from fastapi import FastAPI
- from motor.motor_asyncio import AsyncIOMotorClient
- from models import Student

- app = FastAPI()

- # Mongo connection
- client = AsyncIOMotorClient("mongodb://localhost:27017")
- db = client.school

- @app.post("/students/")
- async def create(s: Student):
- await db.students.insert_one(s.model_dump())
- return {"status": "created"}

- @app.get("/students/{student_id}")
- async def read(student_id: str):
- doc = await db.students.find_one({"student_id": student_id})
- if doc:
- doc["_id"] = str(doc["_id"])
- return doc

- @app.delete("/students/{student_id}")
- async def delete(student_id: str):
- await db.students.delete_one({"student_id": student_id})
- return {"status": "deleted"}

Step 8: Run FastAPI Server

```
# run  
uvicorn main:app --reload
```

```
# u must to ssee  
http://127.0.0.1:8000
```

Step 9: Test API (POST)

run the following to CREATE

```
curl -X POST http://localhost:8000/students/  
-H 'Content-Type: application/json'  
-d '{"student_id":"101","name":"John","grade":3}'
```

Step 10: Test API (GET)

run to READ

```
curl http://localhost:8000/students/101
```

Step 11: Test API (DELETE)

run to DELETE

```
curl -X DELETE http://localhost:8000/students/101
```

Common Issues

#Check if server runs

```
curl http://127.0.0.1:8000/docs
```

#Check ports

```
ss -tulnp | grep 8000
```

#Check Docker

```
sudo systemctl status docker
```

```
docker ps
```